

Liponic® EG-1

10007223

An Ideal Multifunctional Ingredient



PRODUCT: Liponic® EG-1

EINECS #: Exempt

INCI NAME: Glycereth-26

CAS #: 31694-55-0

WHAT IS LIPONIC® EG-1?

Liponic® EG-1 is the 26 mole ethoxylate of glycerin. A multifunctional ingredient that can be used as a humectant, solubilizer, co-emulsifier, and skin conditioning agent. Additionally, it may be utilized as a butylene glycol replacement. It functions as a dispersant, detackifier, solubilizer, viscosity modifier, plasticizer, foam modifier and an exceptional lubricant.

WHAT DOES LIPONIC® EG-1 DO?

In skin care applications, **Liponic® EG-1** provides superior skin moisturization through its humectant property and film forming on the skin. Skin conductance measurements using the SkinLab™ instrument document this moisturization performance based on a ten person panel test applying a lotion with 5% **Liponic® EG-1**. Additional key benefits provided by **Liponic® EG-1** include the reduction of playtime, smooth and soft skin feel, elegant and luxurious appearance of emulsion systems, and excellent solubilization properties for colorants, powders and active ingredients.

Liponic® EG-1 provides a smooth elegant feel without the tackiness normally associated with glycerin. If used in combination with glycerin or butylene glycol, it will help reduce the tackiness and improve the overall skin feel, providing elegance to formulations.

Liponic® EG-1 also demonstrates performance benefits in cleansing surfactant systems boosting foam volume and foam density. Numerous studies were designed and conducted to document these improvements in a sulfate-free body wash and a sulfate-free shampoo.

KEY BENEFITS

- Long Lasting Skin Moisturization
- Imparts a Non-Tacky, Smooth & Luxurious Skin Feel
- Creates Bright White Elegant Formulations
- Improves Foam Volume & Density
- Conditioned Feel on Hair
- Solubilization Properties for Colorants, Powders & Active Ingredients

IDEAL FOR USE

- Creams & Lotions
- Suncare Products
- BB Creams & CC Creams
- Facial Masks & Cleansers
- Body Washes & Liquid Soaps
- Shampoos

 Vantage

TYPICAL PROPERTIES OF LIPONIC® EG-1

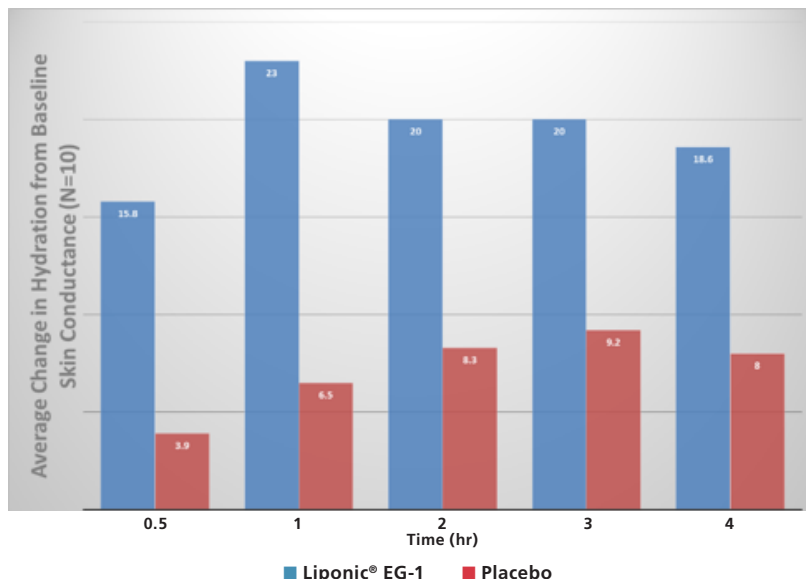
Appearance @ 25° C	Clear to hazy, slightly viscous liquid
Odor	Characteristic
Color APHA (50% aq. solution)	30 max
Moisture Content, %	1 max
Solubility	Soluble in water and alcohol; Insoluble in mineral and vegetable oils

HUMECTANTS

SKIN MOISTURIZATION BENEFITS OF LIPONIC® EG-1

Liponic® EG-1 delivered by an oil-in-water lotion provided superior moisturization vs. placebo lotion.

Moisture readings were taken with the SkinLab™ moisture probe measuring conductance. Skin baseline readings were taken on the forearms of 10 panelists and then a 'control' lotion and a lotion with 5% **Liponic® EG-1** were applied to separate forearms to measure changes over baseline. Readings were taken at 1 hour intervals and showed enhanced moisturization using **Liponic® EG-1** when compared to the control.



IMPROVEMENT OF FOAM QUALITY IN SULFATE-FREE SURFACTANT SYSTEMS USING LIPONIC® EG-1

The image demonstrates improved foam volume and foam density of **Liponic® EG-1** in a sulfate-free cleansing surfactant system. Sulfate-free cleansing surfactant systems (shampoo, body wash and facial) are known to have low foaming and a loose foam structure. A graduated cylinder test was used to conduct a side-by-side evaluation to show visual performance gains. Glycerin is typically used in these systems to improve skin-feel however, it does not result in good foam structure.

Glycerin vs LIPONIC® EG-1

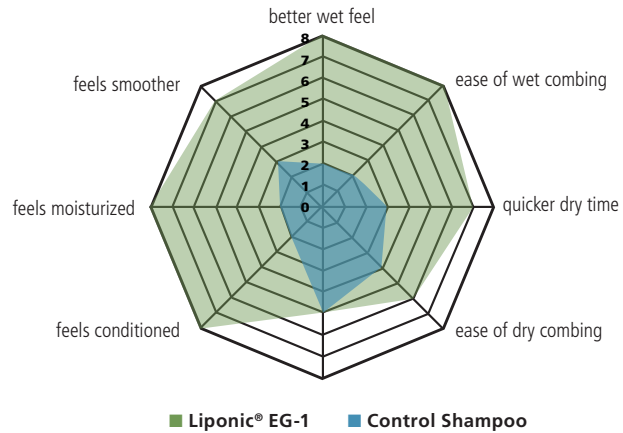


3 minutes after placement

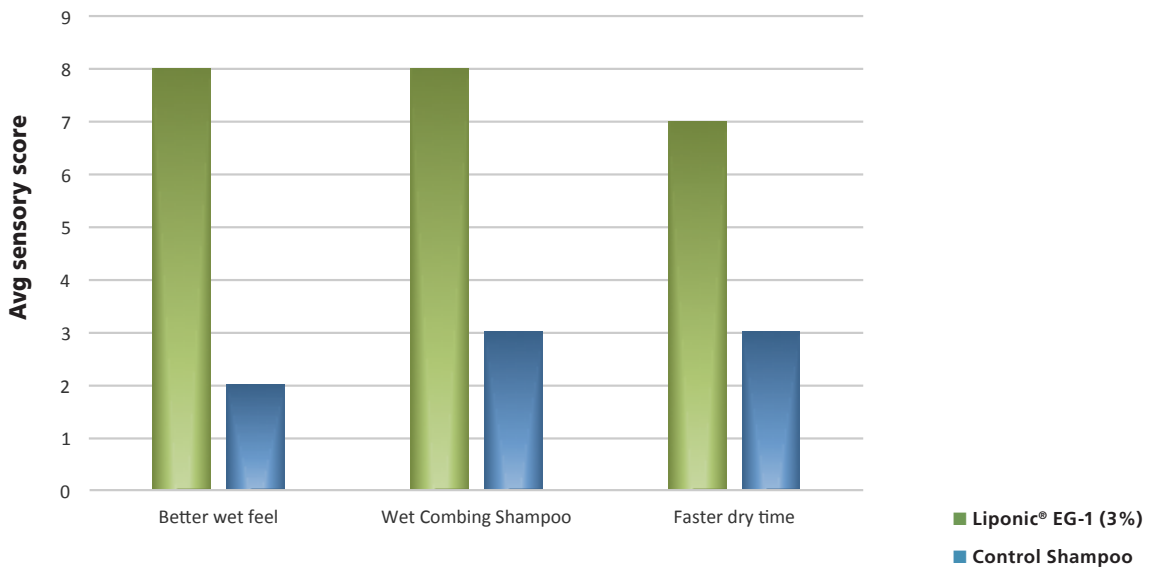


HAIR TRESS STUDY OF LIPONIC® EG-1 IN A SULFATE-FREE SURFACTANT SYSTEM

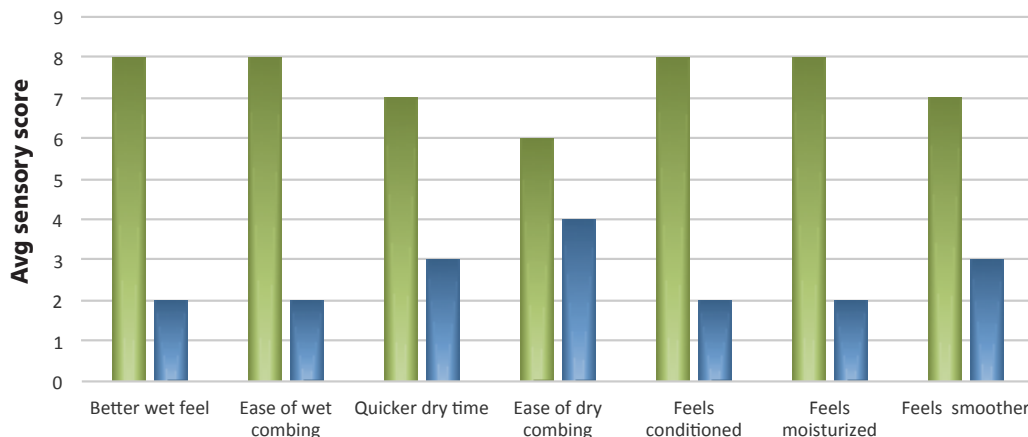
Hair tress testing is another key parameter to further evaluate the improved performance of **Liponic® EG-1** in a surfactant-free shampoo. Panelists evaluated wet and dry combing, hair feel and dry time properties of formulations with 3% **Liponic® EG-1** in comparison to a control without **Liponic® EG-1**.



Wet Performance Properties in Sulfate-Free Shampoo

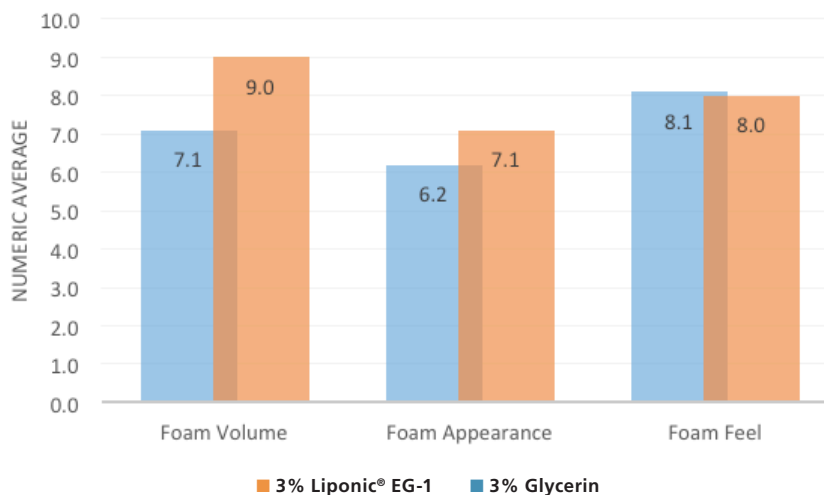


Dry Performance Properties in Sulfate-Free Shampoo



BODY WASH PANEL TEST USING LIPONIC® EG-1

Hand washing studies were performed to demonstrate perceived improvement in foam volume and foam appearance benefits. **Liponic® EG-1** was placed in a sulfate-free body wash formulation and compared to glycerin in the same formulation base.



HOW CAN LIPONIC® EG-1 BE USED?

Liponic® EG-1 can be used in a variety of applications. Use levels range from 0.5 to 10%. It is ideal for use in emulsion systems, surfactant systems, color cosmetics and facial masks.

Liponic® EG-1 is water soluble and can be used in both hot and cold process formulations.



All data, including the formulations and procedures discussed herein, to the knowledge of Vantage, are believed to be correct, reliable and accurate. Please note, however, that Vantage does not warrant or guarantee any accuracy, reliability or completeness of the information contained herein. It is the user's responsibility to determine the suitability and completeness of such information for the user's particular use (including performing any necessary confirmatory tests). Vantage is not responsible or liable for any loss or damage that may occur from the use of this information, nor do we warrant against any patent infringement. Nothing contained herein shall be construed as providing any permission, recommendation, or inducement to practice any patented invention without permission of the patent owner.